07. Statistics & Averages

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Finding the mean:

Find the mean of 5, 9, 7, 11, 3.

Answer: Add them all up: 5 + 9 + 7 + 11 + 3 = 35. Divide by how many numbers there are: 35 ÷ 5 = **7**.

Example 2 — Finding the mode and the median:

Find the mode and the median of 6, 2, 9, 2, 8.

Answer: Order the numbers first: 2, 2, 6, 8, 9. Mode = the value that appears most often = **2**. Median = the middle value = **6**.

Example 3 — Mean and range from a table:

Sweets sold over 4 days: Mon = 16, Tue = 24, Wed = 20, Thu = 12. Find the mean sold per day and the range.

Answer: Mean: (16 + 24 + 20 + 12) ÷ 4 = 72 ÷ 4 = **18**. Range: 24 − 12 = **12**.

Example 4 — Reading a frequency table:

A frequency table shows: 1 pet — 7 pupils, 2 pets — 10 pupils, 3 pets — 4 pupils. How many pupils were surveyed in total, and what is the mode number of pets owned?

Answer: Total surveyed: 7 + 10 + 4 = **21 pupils**. The mode is the most common value in the table, which is the one with the highest frequency (10 pupils) — the mode is **2 pets**.

Example 5 — Multi-step: pie chart proportions:

In a class of 24 pupils, a pie chart shows that half walk to school and a third cycle. Everyone else goes by car. How many pupils go by car?

Answer: Half of 24 = 12 walk. A third of 24 = 8 cycle. Walk + cycle = 12 + 8 = 20. The rest go by car: 24 − 20 = **4 pupils**.

Foundation

15 questions

1. Find the mean of 4, 6, 8, 10, 2.

2. Find the mean of 9, 15, 12.

3. Find the median of 5, 1, 8, 3, 9.

4. Find the median of 10, 2, 16, 6, 12.

5. Find the mode of 4, 7, 7, 9, 3, 7.

6. Find the mode of 6, 6, 8, 8, 8, 2.

7. Find the range of 5, 13, 8, 19, 10.

8. Find the range of 40, 25, 55, 20.

9. Which of these is the mode of 3, 3, 5, 7, 7, 7? A) 7 B) 3 C) 5 D) 6

10. A table shows fruit sold: Apples = 9, Bananas = 6, Pears = 4, Oranges = 5. How many pieces of fruit in total?

11. Sweets sold on 4 days: Mon = 10, Tue = 16, Wed = 8, Thu = 14. What is the mean number of sweets sold per day?

12. Select ALL that apply. A frequency table shows: 1 pet = 5 pupils, 2 pets = 8 pupils, 3 pets = 4 pupils. Which statements are true? A) 17 pupils were surveyed in total B) The mode number of pets owned is 2 C) The mode number of pets owned is 1 D) More pupils own 2 pets than own 3 pets

13. Four friends scored 12, 16, 18, 14 in a quiz. What is their mean score? A) 15 B) 16 C) 17 D) 14

14. What is the mean of 4, 4, 4, 4, 9? A) 5 B) 4 C) 6 D) 7

15. Find the range of 7, 18, 13, 2, 10.

Standard

20 questions

1. Find the median of 3, 8, 1, 6.

2. Find the mean of 14, 18, 10, 22.

3. A pictogram shows books read: Ella = 5, Finn = 8, Gia = 2, Hugo = 11. How many books were read altogether?

4. Which of these is the median of 4, 11, 7, 19, 2? A) 7 B) 11 C) 4 D) 19

5. Find the range of 16, 38, 23, 7, 31.

6. A bar chart shows sales at 4 shops: A = 13, B = 20, C = 7, D = 16. What is the mean sale per shop?

7. Find the mode of 3, 5, 5, 7, 9, 5.

8. A table shows pets: Dogs = 10, Cats = 7, Fish = 5, Birds = 2. If 6 more dogs are added, how many pets are there altogether now?

9. Select ALL that apply. A class recorded favourite colours: Blue = 9, Red = 5, Green = 3. Which statements are true? A) 17 pupils were asked in total B) Blue is the mode (most popular) C) Green is the mode (most popular) D) More pupils chose Red than Blue

10. Find the mean of 6, 10, 8, 12.

11. Which of these is the range of 29, 11, 24, 38, 16? A) 27 B) 38 C) 11 D) 18

12. Find the median of 5, 2, 8, 3, 10, 6.

13. A line graph shows temperatures (°C) over 4 days: 9, 14, 10, 18. What is the range of the temperatures?

14. Which of these is the mean of 8, 12, 16, 20? A) 14 B) 16 C) 12 D) 18

15. Find the range of 42, 60, 55, 47.

16. Over five days, a shop's sales figures were 18, 12, 22, 8 and 25. What is the mean of these five numbers?

17. Five numbers are 14, 32, 5, 21 and 27. Once they are placed in order, what is the median?

18. In a class of 28 pupils, a pie chart shows that half walk to school and a quarter cycle. Everyone else arrives by car. How many pupils is that?

19. Find the mean of 20, 30, 25, 15, 10.

20. A frequency table shows exam scores: 5 pupils scored 60, 8 pupils scored 70, 3 pupils scored 80. How many pupils took the exam in total?

Hard

5 questions

1. Six numbers have a mean of 9. Five of the numbers are 7, 10, 6, 12 and 8. What is the sixth number?

2. The mean of four numbers is 15. Three of the numbers are 12, 18 and 9. What is the fourth number?

3. A class of 32 pupils is surveyed about pets. Half own a dog, a quarter own a cat, and the rest own neither. How many pupils own neither?

4. Five test scores are 60, 75, 80, 65 and 70. Select ALL the statements below that are TRUE: A) The mean is 70 B) The median is 70 C) The range is 20 D) There is a mode, because one score appears more than once

5. A bag of 7 numbers has a mean of 12. One of the numbers, 18, is removed. What is the mean of the remaining 6 numbers?

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